



Universidad Distrital Francisco José de Caldas

Facultad de Ingeniería
Ingeniería Catastral y Geodesia

Entrega 1
Curso Data Science ACM UD

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1. Desarrollo

Se implementó una base de datos relacional en MySQL que contiene las tablas `award`, `award_winner`, `city`, `confederation`, `country`, `federation`, `goal`, `matches`, `player`, `player_appearance`, `position`, `region`, `stadium`, `team` y `tournament`, todas con clave primaria, claves foráneas cuando corresponde, tipos de datos adecuados e integridad referencial.

Además, se desarrollaron scripts en Python que leen los archivos CSV con Pandas, establecen la conexión con MySQL, insertan la información en las tablas y muestran mensajes de carga exitosa, partiendo de datos limpios.

Visión general del procedimiento

El trabajo se organizó en cuatro etapas consecutivas, primero se limpian los archivos CSV, luego se crea la base de datos en el servidor luego de hacer las respectivas configuraciones en XAMPP, después se cargan los archivos con Python (lo que genera un esquema *plano*, sin restricciones) y, finalmente, se construye a partir de ese esquema plano un esquema *normalizado* con claves primarias, claves foráneas e integridad referencial.

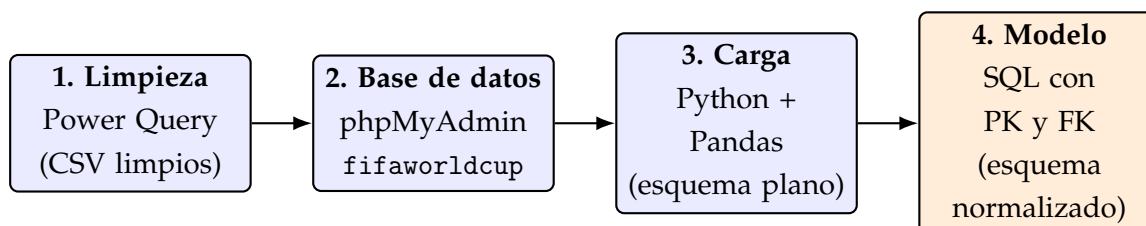


Figura 1: Flujo de trabajo del Avance 1

1. Limpieza y estandarización de los archivos CSV

Antes de cargar los datos se depuraron los archivos CSV con el editor de Power Query de Excel. Se hicieron dos correcciones.

Valores no aplicables. Los archivos originales usaban el texto *not applicable* para indicar que un dato no corresponde a un registro pero ese texto no es un valor válido para columnas numéricas, de fecha o booleanas, entonces se reemplazó por NULL.

Formato de fecha y hora. El formato que exporta al CSV puede variar (por ejemplo, dd/mm/aaaa) y MySQL reconoce sin ambigüedad los formatos YYYY-MM-DD para fechas y HH:MM:SS para horas. Por eso las columnas `match_date` y `match_time` se convirtieron a texto con ese formato, de modo que después puedan pasarse a los tipos DATE y TIME.

En *Agregar columna* → *Columna personalizada*, crear la columna `match_date_nueva` con la fórmula:

```
Date.ToText([match_date], "yyyy-MM-dd")
```

Listing 1: Fórmula de Power Query para la fecha

Y Crear la columna personalizada `match_time_nueva` con la fórmula:

```
Time.ToText([match_time], "HH:mm:ss")
```

Listing 2: Fórmula de Power Query para la hora

2. Creación de la base de datos en phpMyAdmin

Con los archivos limpios se creó en phpMyAdmin, sobre el servidor MySQL local, una base de datos vacía llamada `fifaworldcup` que recibe los 27 archivos CSV tal como vienen, sin restricciones.

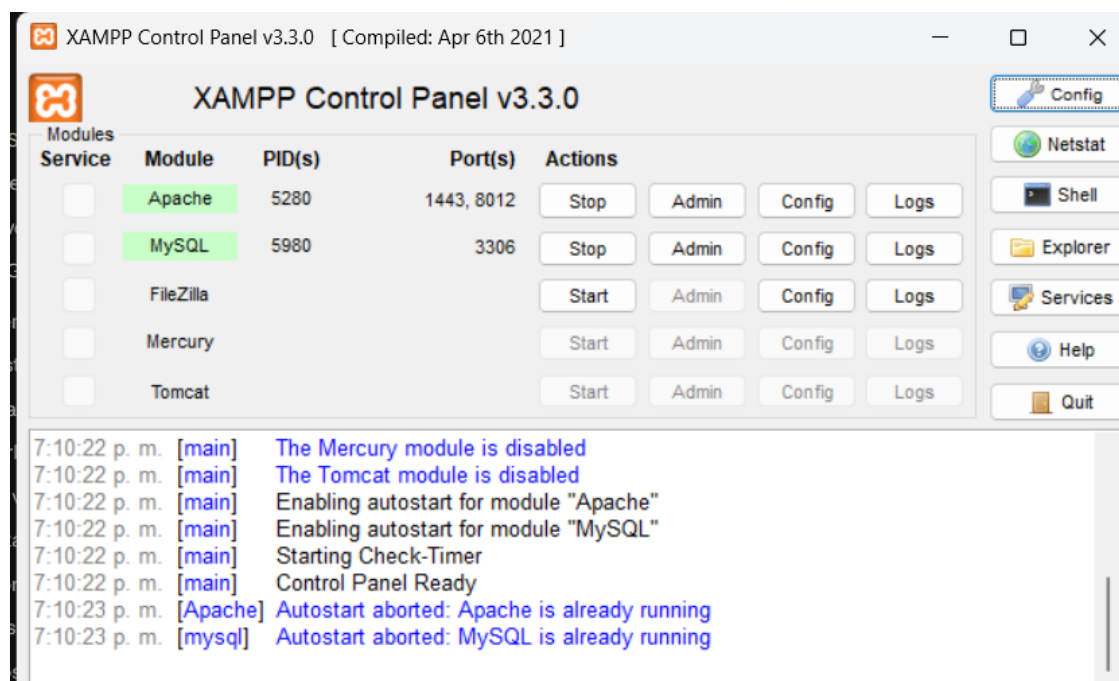


Figura 2: Cpnfiguración XAMPP

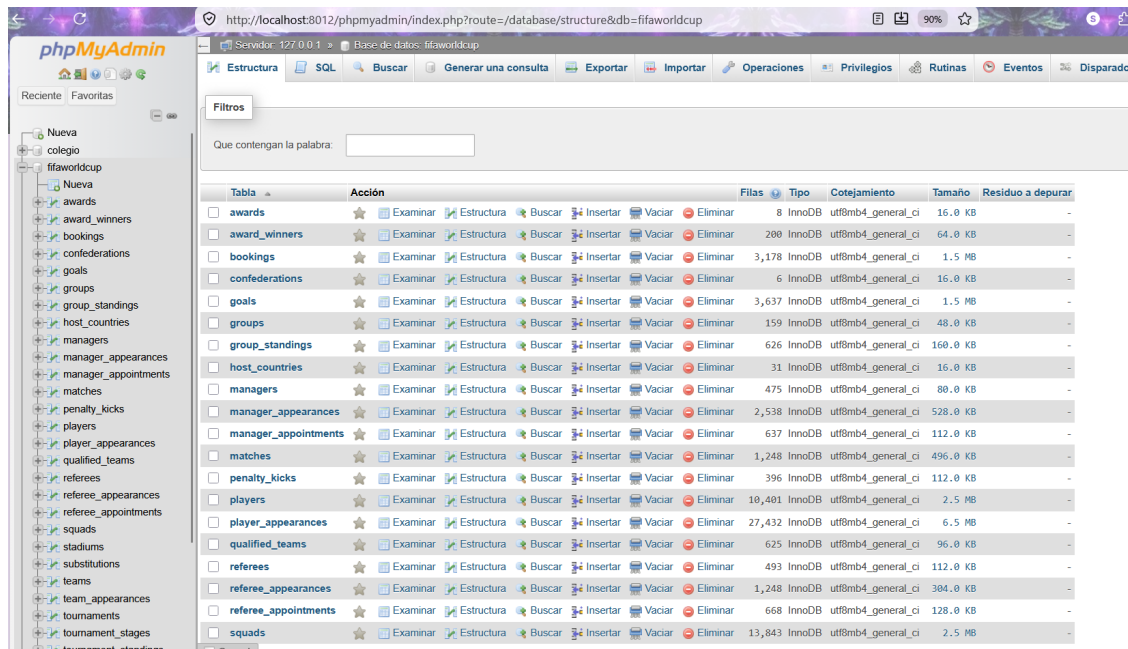


Figura 3: Base de datos *fifaworldcup* creada en phpMyAdmin

3. Carga de los archivos CSV con Python

La carga se automatizó con dos scripts: uno que define la conexión y otro que recorre la carpeta de CSV y sube cada archivo a MySQL.

Conexión a MySQL (`connection.py`)

Este módulo define la función `get_connection_engine`, que construye una cadena de conexión con el controlador `pymysql` y crea un motor de `SQLAlchemy` hacia la base `fifaworldcup`. El decorador `@st.cache_resource` de `Streamlit` guarda el motor en memoria para reutilizarlo en lugar de abrir una conexión nueva en cada ejecución.

```

1 import streamlit as st
2 from sqlalchemy import create_engine
3
4 USER = "root"
5 PASSWORD = ""
6 HOST = "localhost"
7 PORT = 3306
8 DATABASE = "fifaworldcup"
9

```

```

10 @st.cache_resource
11 def get_connection_engine():
12     """Crea y cachea el motor de conexión a la base de datos de MySQL
        """
13     connection_string = f"mysql+pymysql://{USER}:{PASSWORD}@{HOST}:{
        PORT}/{DATABASE}"
14     engine = create_engine(connection_string)
15     return engine

```

Listing 3: connection.py

Carga de los archivos (1_cargarcsv.py)

El script recorre la carpeta csvsubir y, por cada archivo con extensión .csv, realiza los siguientes pasos:

- **Nombre de la tabla:** se toma del nombre del archivo, sin extensión y en minúsculas (`os.path.splitext(...)[0].lower()`).
- **Lectura con Pandas:** `pd.read_csv` con `sep=";"`, porque los archivos usan punto y coma como separador, y `encoding="utf-8-sig"`, que elimina la marca BOM al inicio del archivo y conserva las tildes y caracteres especiales de los nombres.
- **Inserción:** `df.to_sql(tabla, engine, if_exists="replace", index=False)` crea la tabla e inserta todas las filas. Con `replace`, si la tabla ya existe se elimina y se vuelve a crear, lo que hace la carga repetible. Con `index=False` no se guarda el índice de Pandas como columna.
- **Mensaje de control:** `print(tabla, len(df))` muestra el nombre de la tabla y el número de filas cargadas, y confirma que la carga fue exitosa.

```

1 import os
2 import pandas as pd
3 from sqlalchemy import create_engine
4
5 DATABASE = "fifaworldcup"
6 CARPETA = r"C:\Users\Usuario\Documents\DataSci\datos\csvsubir"
7

```

```

8 engine = create_engine(f"mysql+pymysql://root:@localhost:3306/{
    DATABASE}")
9
10 for archivo in os.listdir(CARPETA):
11     if archivo.lower().endswith(".csv"):
12         tabla = os.path.splitext(archivo)[0].lower()
13         df = pd.read_csv(os.path.join(CARPETA, archivo), sep=";",
14                             encoding="utf-8-sig")
15         df.to_sql(tabla, engine, if_exists="replace", index=False)
16         print(tabla, len(df))

```

Listing 4: 1_cargarcsv.py

Resumen de tablas, claves y relaciones

Tabla	Clave primaria	Claves foráneas	UNIQUE
confederation	confederation_id	—	—
region	region_id	confederation_id → confederation	→ (region_name, confederation_id)
country	country_id	region_id → region	country_name
federation	federation_id	confederation_id → confederation	→ federation_name
team	team_id	federation_id → federation country_id → country	—
city	city_id	country_id → country	(city_name, country_id)
stadium	stadium_id	city_id → city	—
tournament	tournament_id	host_country_id → country winner_team_id → team	—
matches	match_id	tournament_id → tournament stadium_id → stadium home_team_id → team away_team_id → team	—
position	position_id	—	position_name
player	player_id	—	—

Tabla	Clave primaria	Claves foráneas	UNIQUE
player_appearance	player_appearance_id	match_id → matches player_id → player team_id → team position_id → position	(match_id, player_id)
goal	goal_id	match_id → matches player_id → player team_id → team	—
award	award_id	—	—
award_winner	award_winner_id	tournament_id → tournament award_id → award player_id → player team_id → team	—

Script SQL

El script se presenta en cinco bloques, en el mismo orden en que se ejecuta. La numeración de líneas coincide con la del archivo original.

Bloque 1: esquema, confederaciones, regiones, países y federaciones. Se crea el esquema y las tablas de base que no dependen de otras (confederation), seguidas de las que dependen de ella.

```

1  -- =====
2  -- Normalización FIFA World Cup
3  -- Lee datos desde el esquema plano `fifaworldcup` (ya cargado con
4  -- 1_cargarcsv.py) y construye el esquema normalizado
5  -- `fifaworldcup_normalized` con las 15 tablas pedidas.
6  -- Se usa un esquema nuevo porque el origen ya tiene una tabla llamada
7  -- `matches`; así no se pisa el dato crudo que se necesita para poblar.
8  -- =====
9
10 DROP DATABASE IF EXISTS fifaworldcup_normalized;
11 CREATE DATABASE fifaworldcup_normalized CHARACTER SET utf8mb4;
12 USE fifaworldcup_normalized;
13
14 -- -----
15 -- confederation
16 -- -----
17 CREATE TABLE confederation (
18     confederation_id VARCHAR(10) NOT NULL,
19     confederation_name VARCHAR(150) NOT NULL,

```

```

20     confederation_code VARCHAR(10),
21     confederation_wikipedia_link VARCHAR(255),
22     PRIMARY KEY (confederation_id)
23 ) ENGINE=InnoDB;
24
25 INSERT INTO confederation (confederation_id, confederation_name,
26     confederation_code, confederation_wikipedia_link)
27 SELECT confederation_id, MAX(confederation_name), MAX(confederation_code),
28     MAX(confederation_wikipedia_link)
29 FROM fifaworldcup.confederations
30 WHERE confederation_id IS NOT NULL
31 GROUP BY confederation_id;
32
33 -----
34 -- region
35 -----
36 CREATE TABLE region (
37     region_id INT NOT NULL AUTO_INCREMENT,
38     region_name VARCHAR(100) NOT NULL,
39     confederation_id VARCHAR(10),
40     PRIMARY KEY (region_id),
41     UNIQUE KEY uq_region (region_name, confederation_id),
42     CONSTRAINT fk_region_confederation FOREIGN KEY (confederation_id)
43     REFERENCES confederation (confederation_id)
44 ) ENGINE=InnoDB;
45
46 INSERT INTO region (region_name, confederation_id)
47 SELECT DISTINCT t.region_name, t.confederation_id
48 FROM fifaworldcup.teams t
49 WHERE t.region_name IS NOT NULL AND t.region_name <> '';
50
51 -----
52 -- country
53 -----
54 CREATE TABLE country (
55     country_id INT NOT NULL AUTO_INCREMENT,
56     country_name VARCHAR(150) NOT NULL,
57     region_id INT,
58     PRIMARY KEY (country_id),
59     UNIQUE KEY uq_country_name (country_name),
60     CONSTRAINT fk_country_region FOREIGN KEY (region_id)
61     REFERENCES region (region_id)
62 ) ENGINE=InnoDB;
63
64 INSERT INTO country (country_name)

```

```

63 SELECT DISTINCT country_name FROM (
64     SELECT country_name FROM fifaworldcup.stadiums
65     UNION
66     SELECT country_name FROM fifaworldcup.matches
67     UNION
68     SELECT host_country AS country_name FROM fifaworldcup.tournaments
69     UNION
70     SELECT team_name AS country_name FROM fifaworldcup.teams
71 ) src
72 WHERE country_name IS NOT NULL AND country_name <> '';
73
74 UPDATE country c
75 JOIN fifaworldcup.teams t ON t.team_name = c.country_name
76 JOIN region r ON r.region_name = t.region_name AND r.confederation_id = t.
    confederation_id
77 SET c.region_id = r.region_id;
78
79 -----
80 -- federation
81 -----
82 CREATE TABLE federation (
83     federation_id INT NOT NULL AUTO_INCREMENT,
84     federation_name VARCHAR(150) NOT NULL,
85     federation_wikipedia_link VARCHAR(255),
86     confederation_id VARCHAR(10),
87     PRIMARY KEY (federation_id),
88     UNIQUE KEY uq_federation_name (federation_name),
89     CONSTRAINT fk_federation_confederation FOREIGN KEY (confederation_id)
90     REFERENCES confederation (confederation_id)
91 ) ENGINE=InnoDB;
92
93 INSERT INTO federation (federation_name, federation_wikipedia_link,
    confederation_id)
94 SELECT federation_name, MAX(federation_wikipedia_link), MAX(
    confederation_id)
95 FROM fifaworldcup.teams
96 WHERE federation_name IS NOT NULL AND federation_name <> ''
97 GROUP BY federation_name;

```

Listing 5: Bloque 1 de código PK_FK

Bloque 2: equipos, ciudades y estadios. Cada equipo se enlaza con su federación y con su país. Las ciudades se extraen de la tabla de estadios y cada estadio pasa a apuntar a su ciudad.

```

99  -- -----
100 -- team
101  -- -----
102 CREATE TABLE team (
103     team_id VARCHAR(10) NOT NULL,
104     team_name VARCHAR(100) NOT NULL,
105     team_code VARCHAR(10),
106     mens_team TINYINT(1) DEFAULT 0,
107     womens_team TINYINT(1) DEFAULT 0,
108     federation_id INT,
109     country_id INT,
110     mens_team_wikipedia_link VARCHAR(255),
111     womens_team_wikipedia_link VARCHAR(255),
112     PRIMARY KEY (team_id),
113     CONSTRAINT fk_team_federation FOREIGN KEY (federation_id)
114         REFERENCES federation (federation_id),
115     CONSTRAINT fk_team_country FOREIGN KEY (country_id)
116         REFERENCES country (country_id)
117 ) ENGINE=InnoDB;
118
119 INSERT INTO team (team_id, team_name, team_code, mens_team, womens_team,
120     federation_id, country_id,
121     mens_team_wikipedia_link, womens_team_wikipedia_link)
122 SELECT t.team_id, t.team_name, t.team_code, t.mens_team, t.womens_team,
123     f.federation_id, c.country_id,
124     t.mens_team_wikipedia_link, t.womens_team_wikipedia_link
125 FROM fifaworldcup.teams t
126 LEFT JOIN federation f ON f.federation_name = t.federation_name
127 LEFT JOIN country c ON c.country_name = t.team_name;
128
129 -- -----
130 -- city
131 -- -----
132 CREATE TABLE city (
133     city_id INT NOT NULL AUTO_INCREMENT,
134     city_name VARCHAR(150) NOT NULL,
135     country_id INT,
136     city_wikipedia_link VARCHAR(255),
137     PRIMARY KEY (city_id),
138     UNIQUE KEY uq_city (city_name, country_id),
139     CONSTRAINT fk_city_country FOREIGN KEY (country_id)
140         REFERENCES country (country_id)
141 ) ENGINE=InnoDB;
142 INSERT INTO city (city_name, country_id, city_wikipedia_link)

```

```

143 SELECT s.city_name, c.country_id, MAX(s.city_wikipedia_link)
144 FROM fifaworldcup.stadiums s
145 LEFT JOIN country c ON c.country_name = s.country_name
146 WHERE s.city_name IS NOT NULL AND s.city_name <> ''
147 GROUP BY s.city_name, c.country_id;
148
149 -----
150 -- stadium
151 -----
152 CREATE TABLE stadium (
153     stadium_id VARCHAR(10) NOT NULL,
154     stadium_name VARCHAR(150) NOT NULL,
155     city_id INT,
156     stadium_capacity INT,
157     stadium_wikipedia_link VARCHAR(255),
158     PRIMARY KEY (stadium_id),
159     CONSTRAINT fk_stadium_city FOREIGN KEY (city_id)
160         REFERENCES city (city_id)
161 ) ENGINE=InnoDB;
162
163 INSERT INTO stadium (stadium_id, stadium_name, city_id, stadium_capacity,
164     stadium_wikipedia_link)
165 SELECT s.stadium_id, s.stadium_name, ci.city_id, s.stadium_capacity, s.
166     stadium_wikipedia_link
167 FROM fifaworldcup.stadiums s
168 LEFT JOIN country co ON co.country_name = s.country_name
169 LEFT JOIN city ci ON ci.city_name = s.city_name
170 AND (ci.country_id = co.country_id OR (ci.country_id IS NULL AND co.
171     country_id IS NULL));

```

Listing 6: Bloque 2 de código PK_FK

Bloque 3: torneos y partidos. El torneo referencia al país anfitrión y al equipo ganador; el partido referencia al torneo, al estadio y a los equipos local y visitante.

```

170 -----
171 -- tournament
172 -----
173 CREATE TABLE tournament (
174     tournament_id VARCHAR(10) NOT NULL,
175     tournament_name VARCHAR(150) NOT NULL,
176     `year` YEAR,
177     start_date DATE,
178     end_date DATE,
179     host_country_id INT,

```

```

180 winner_team_id VARCHAR(10),
181 host_won TINYINT(1),
182 count_teams INT,
183 group_stage TINYINT(1),
184 second_group_stage TINYINT(1),
185 final_round TINYINT(1),
186 round_of_16 TINYINT(1),
187 quarter_finals TINYINT(1),
188 semi_finals TINYINT(1),
189 third_place_match TINYINT(1),
190 `final` TINYINT(1),
191 PRIMARY KEY (tournament_id),
192 CONSTRAINT fk_tournament_host FOREIGN KEY (host_country_id)
193 REFERENCES country (country_id),
194 CONSTRAINT fk_tournament_winner FOREIGN KEY (winner_team_id)
195 REFERENCES team (team_id)
196 ) ENGINE=InnoDB;
197
198 INSERT INTO tournament (tournament_id, tournament_name, `year`, start_date
199 , end_date, host_country_id,
200 winner_team_id, host_won, count_teams,
201 group_stage, second_group_stage,
202 final_round, round_of_16, quarter_finals,
203 semi_finals, third_place_match, `final`)
204 SELECT tr.tournament_id, tr.tournament_name, tr.`year`, NULLIF(tr.
205 start_date, ''), NULLIF(tr.end_date, ''),
206 co.country_id, tm.team_id,
207 tr.host_won, tr.count_teams, tr.group_stage, tr.second_group_stage,
208 tr.final_round,
209 tr.round_of_16, tr.quarter_finals, tr.semi_finals, tr.
210 third_place_match, tr.`final`
211 FROM fifaworldcup.tournaments tr
212 LEFT JOIN country co ON co.country_name = tr.host_country
213 LEFT JOIN team tm ON tm.team_name = tr.winner;
214
215 -----
216 -- matches
217 -----
218 CREATE TABLE matches (
219 match_id VARCHAR(20) NOT NULL,
220 tournament_id VARCHAR(10),
221 match_name VARCHAR(150),
222 stage_name VARCHAR(50),
223 group_name VARCHAR(20),
224 group_stage TINYINT(1),

```

```

219     knockout_stage TINYINT(1),
220     match_date DATE,
221     match_time TIME,
222     stadium_id VARCHAR(10),
223     home_team_id VARCHAR(10),
224     away_team_id VARCHAR(10),
225     home_team_score INT,
226     away_team_score INT,
227     extra_time TINYINT(1),
228     penalty_shootout TINYINT(1),
229     home_team_score_penalties INT,
230     away_team_score_penalties INT,
231     result VARCHAR(50),
232     PRIMARY KEY (match_id),
233     CONSTRAINT fk_match_tournament FOREIGN KEY (tournament_id)
234         REFERENCES tournament (tournament_id),
235     CONSTRAINT fk_match_stadium FOREIGN KEY (stadium_id)
236         REFERENCES stadium (stadium_id),
237     CONSTRAINT fk_match_home_team FOREIGN KEY (home_team_id)
238         REFERENCES team (team_id),
239     CONSTRAINT fk_match_away_team FOREIGN KEY (away_team_id)
240         REFERENCES team (team_id)
241 ) ENGINE=InnoDB;
242
243 INSERT INTO matches (match_id, tournament_id, match_name, stage_name,
244                     group_name, group_stage,
245                     knockout_stage, match_date, match_time, stadium_id,
246                     home_team_id, away_team_id,
247                     home_team_score, away_team_score, extra_time,
248                     penalty_shootout,
249                     home_team_score_penalties, away_team_score_penalties
250                     , result)
251 SELECT match_id, tournament_id, match_name, stage_name, group_name,
252        group_stage,
253        knockout_stage, NULLIF(match_date, ''), NULLIF(match_time, ''),
254        stadium_id, home_team_id, away_team_id,
255        home_team_score, away_team_score, extra_time, penalty_shootout,
256        home_team_score_penalties, away_team_score_penalties, result
257 FROM fifaworldcup.matches;

```

Listing 7: Bloque 3 de código PK_FK

Bloque 4: posiciones, jugadores y apariciones. La tabla `player_appearance` relaciona a un jugador con un partido, un equipo y una posición.

```

253  -----
254  -- position
255  -----
256  CREATE TABLE `position` (
257      position_id INT NOT NULL AUTO_INCREMENT,
258      position_name VARCHAR(50) NOT NULL,
259      position_code VARCHAR(5),
260      PRIMARY KEY (position_id),
261      UNIQUE KEY uq_position_name (position_name)
262  ) ENGINE=InnoDB;
263
264  INSERT INTO `position` (position_name, position_code)
265  SELECT position_name, MAX(position_code) FROM (
266      SELECT position_name, position_code FROM fifaworldcup.player_appearances
267      UNION ALL
268      SELECT position_name, position_code FROM fifaworldcup.squads
269  ) p
270  WHERE position_name IS NOT NULL AND position_name <> ''
271  GROUP BY position_name;
272
273  -----
274  -- player
275  -----
276  CREATE TABLE player (
277      player_id VARCHAR(15) NOT NULL,
278      family_name VARCHAR(100),
279      given_name VARCHAR(100),
280      birth_date DATE,
281      female TINYINT(1),
282      goal_keeper TINYINT(1),
283      defender TINYINT(1),
284      midfielder TINYINT(1),
285      forward TINYINT(1),
286      count_tournaments INT,
287      list_tournaments VARCHAR(255),
288      player_wikipedia_link VARCHAR(255),
289      PRIMARY KEY (player_id)
290  ) ENGINE=InnoDB;
291
292  INSERT INTO player (player_id, family_name, given_name, birth_date, female
293      , goal_keeper, defender,
294      midfielder, forward, count_tournaments,
295      list_tournaments, player_wikipedia_link)
296  SELECT player_id, family_name, given_name, NULLIF(birth_date, ''), female,
297      goal_keeper, defender,

```

```

295         midfielder, forward, count_tournaments, list_tournaments,
           player_wikipedia_link
296 FROM fifaworldcup.players;
297
298 -----
299 -- player_appearance
300 -----
301 CREATE TABLE player_appearance (
302     player_appearance_id INT NOT NULL AUTO_INCREMENT,
303     match_id VARCHAR(20) NOT NULL,
304     player_id VARCHAR(15) NOT NULL,
305     team_id VARCHAR(10),
306     position_id INT,
307     shirt_number INT,
308     starter TINYINT(1),
309     substitute TINYINT(1),
310     PRIMARY KEY (player_appearance_id),
311     UNIQUE KEY uq_match_player (match_id, player_id),
312     CONSTRAINT fk_pa_match FOREIGN KEY (match_id)
313         REFERENCES matches (match_id),
314     CONSTRAINT fk_pa_player FOREIGN KEY (player_id)
315         REFERENCES player (player_id),
316     CONSTRAINT fk_pa_team FOREIGN KEY (team_id)
317         REFERENCES team (team_id),
318     CONSTRAINT fk_pa_position FOREIGN KEY (position_id)
319         REFERENCES `position` (position_id)
320 ) ENGINE=InnoDB;
321
322 INSERT INTO player_appearance (match_id, player_id, team_id, position_id,
           shirt_number, starter, substitute)
323 SELECT pa.match_id, pa.player_id, pa.team_id, po.position_id, pa.
           shirt_number, pa.starter, pa.substitute
324 FROM fifaworldcup.player_appearances pa
325 LEFT JOIN `position` po ON po.position_name = pa.position_name;

```

Listing 8: Bloque 4 de código PK_FK

Bloque 5: goles y premios. Últimas tablas del modelo, que dependen de partidos, jugadores, equipos y torneos.

```

327 -----
328 -- goal
329 -----
330 CREATE TABLE goal (
331     goal_id VARCHAR(10) NOT NULL,

```

```

332     match_id VARCHAR(20),
333     player_id VARCHAR(15),
334     team_id VARCHAR(10),
335     minute_regulation INT,
336     minute_stoppage INT,
337     match_period VARCHAR(50),
338     own_goal TINYINT(1),
339     penalty TINYINT(1),
340     PRIMARY KEY (goal_id),
341     CONSTRAINT fk_goal_match FOREIGN KEY (match_id)
342         REFERENCES matches (match_id),
343     CONSTRAINT fk_goal_player FOREIGN KEY (player_id)
344         REFERENCES player (player_id),
345     CONSTRAINT fk_goal_team FOREIGN KEY (team_id)
346         REFERENCES team (team_id)
347 ) ENGINE=InnoDB;
348
349 INSERT INTO goal (goal_id, match_id, player_id, team_id, minute_regulation
    , minute_stoppage,
350                     match_period, own_goal, penalty)
351 SELECT goal_id, match_id, player_id, player_team_id, minute_regulation,
    minute_stoppage,
352         match_period, own_goal, penalty
353 FROM fifaworldcup.goals;
354
355 -----
356 -- award
357 -----
358 CREATE TABLE award (
359     award_id VARCHAR(10) NOT NULL,
360     award_name VARCHAR(100) NOT NULL,
361     award_description VARCHAR(255),
362     year_introduced YEAR,
363     PRIMARY KEY (award_id)
364 ) ENGINE=InnoDB;
365
366 INSERT INTO award (award_id, award_name, award_description,
    year_introduced)
367 SELECT award_id, award_name, award_description, year_introduced
368 FROM fifaworldcup.awards;
369
370 -----
371 -- award_winner
372 -----
373 CREATE TABLE award_winner (

```

```

374 award_winner_id INT NOT NULL AUTO_INCREMENT,
375 tournament_id VARCHAR(10),
376 award_id VARCHAR(10),
377 shared TINYINT(1),
378 player_id VARCHAR(15),
379 team_id VARCHAR(10),
380 PRIMARY KEY (award_winner_id),
381 CONSTRAINT fk_aw_tournament FOREIGN KEY (tournament_id)
382 REFERENCES tournament (tournament_id),
383 CONSTRAINT fk_aw_award FOREIGN KEY (award_id)
384 REFERENCES award (award_id),
385 CONSTRAINT fk_aw_player FOREIGN KEY (player_id)
386 REFERENCES player (player_id),
387 CONSTRAINT fk_aw_team FOREIGN KEY (team_id)
388 REFERENCES team (team_id)
389 ) ENGINE=InnoDB;
390
391 INSERT INTO award_winner (tournament_id, award_id, shared, player_id,
392 team_id)
393 SELECT tournament_id, award_id, shared, player_id, team_id
394 FROM fifaworldcup.award_winners;

```

Listing 9: Bloque 5 de código PK_FK

Tras ejecutar el script, el esquema `fifaworldcup_normalized` queda con las 15 tablas, sus claves primarias y sus claves foráneas.